

Unit 1.0 – Atomic Structure, Chemical Bonding and Solutions

1.1 Atoms and Fundamental Particles

An **atom** is the smallest particle of an element that can take part in a chemical reaction. It consists of three fundamental subatomic particles:

Particle	Charge	Mass	Location
Electron (e^-)	Negative (-1)	$\sim 1/1837$ amu	Revolves around nucleus
Proton (p^+)	Positive (+1)	1 amu	Inside nucleus
Neutron (n)	Neutral (0)	1 amu	Inside nucleus

1.2 Rutherford Model of Atom

Rutherford, through his gold foil (alpha-particle scattering) experiment, proposed that: - An atom has a small, dense, positively charged **nucleus** at the centre, where almost the entire mass is concentrated. - Electrons revolve around the nucleus in circular orbits. - Most of the atom is empty space.

Limitation: It could not explain the stability of the atom (an orbiting electron should lose energy continuously and spiral into the nucleus), nor the line spectrum of hydrogen.

1.3 Bohr's Theory and Hydrogen Spectrum

Niels Bohr improved Rutherford's model with these postulates: 1. Electrons revolve in fixed circular orbits called **energy levels/shells** without radiating energy. 2. Each orbit has a fixed amount of energy; angular momentum is quantized: $mvr = nh/2\pi$. 3. Energy is absorbed or emitted only when an electron jumps from one orbit to another: $\Delta E = E_2 - E_1 = h\nu$.

Hydrogen spectrum: When an excited electron in hydrogen falls from a higher to a lower energy level, it emits radiation of a specific wavelength, producing spectral lines, grouped into series (Lyman, Balmer, Paschen, etc.) — successfully explained by Bohr's model.

1.4 Wave Mechanical Model, de Broglie and Heisenberg

de Broglie relationship: Matter exhibits dual nature (particle and wave). $\lambda = h / mv$ where λ = wavelength, h = Planck's constant, m = mass, v = velocity.

Heisenberg's Uncertainty Principle: It is impossible to determine simultaneously, with absolute accuracy, both the position and momentum of a microscopic particle like an electron. $\Delta x \cdot \Delta p \geq h / 4\pi$
This principle disproved the concept of fixed orbits and led to the probabilistic (wave-mechanical) model of the atom.

1.5 Quantum Numbers and Shapes of Orbitals

Four quantum numbers describe the position and energy of an electron in an atom: - **Principal quantum number (n):** Shell/energy level ($n = 1, 2, 3, \dots$). - **Azimuthal/Subsidiary quantum number (l):** Sub-

shell/shape of orbital ($l = 0$ to $n-1$); $l=0(s)$, $l=1(p)$, $l=2(d)$, $l=3(f)$. - **Magnetic quantum number (m):** Orientation of orbital in space. - **Spin quantum number (s):** Direction of electron spin ($+1/2$ or $-1/2$).

Shapes of orbitals: s-orbital is spherical; p-orbital is dumb-bell shaped (p_x , p_y , p_z); d-orbital has a double dumb-bell/cloverleaf shape.

1.6 Electronic Configuration

Pauli's Exclusion Principle: No two electrons in an atom can have the same set of all four quantum numbers; an orbital can hold a maximum of 2 electrons with opposite spins.

Hund's Rule of Maximum Multiplicity: Electrons are filled singly into each orbital of a sub-shell before pairing occurs.

Aufbau Principle: Electrons are filled into orbitals in order of increasing energy: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p...

Example — Electronic configuration: - Sodium ($Z=11$): $1s^2 2s^2 2p^6 3s^1$ - Iron ($Z=26$): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

1.7 Chemical Bonding

Cause of chemical bonding: Atoms combine to attain stability by achieving a complete (stable) outer electronic configuration (octet).

Ionic Bond: Formed by complete transfer of electrons from a metal to a non-metal, creating oppositely charged ions held by electrostatic attraction. Example: NaCl ($\text{Na}^+ + \text{Cl}^-$), CaCl_2 , MgO

Covalent Bond: Formed by mutual sharing of electron pairs between atoms. - **Non-polar covalent bond:** Shared equally between identical atoms. Example: H_2 , F_2 - **Polar covalent bond:** Shared unequally between atoms of different electronegativity. Example: HF , HCl

Coordinate (Dative) Bond: Formed when one atom donates both electrons of the shared pair to another atom. Example: CO , NH_4^+ , O_3 , H_2SO_4

1.8 Dipole Moment and Hydrogen Bonding

Dipole moment (μ): A measure of polarity of a bond/molecule = charge $\times$ distance between charges (measured in Debye). Molecules like NH_3 have a net dipole moment due to unsymmetrical shape; NF_3 has a small dipole moment due to opposing bond and lone-pair dipoles.

Hydrogen bonding: A weak electrostatic attraction between a hydrogen atom (bonded to a highly electronegative atom like F, O, N) and another electronegative atom of a nearby molecule. It explains the unusually high boiling point of water.

1.9 Solutions

A **solution** is a homogeneous mixture of two or more substances — **solute** (the substance dissolved, present in smaller quantity) and **solvent** (the dissolving medium, present in larger quantity).

Concentration terms: - **Molarity (M):** Number of moles of solute per litre of solution. $M = \text{moles of solute} / \text{volume of solution (L)}$ - **Normality (N):** Number of gram equivalents of solute per litre of solution. $N = (\text{weight of solute} / \text{equivalent weight}) / \text{volume (L)}$ - **Molality (m):** Number of moles of solute per kilogram of solvent. $m = \text{moles of solute} / \text{mass of solvent (kg)}$

Example: To prepare 1 litre of 1 M NaOH solution, dissolve 40 g of NaOH (molar mass 40 g/mol) in water and make up the volume to 1 litre.

1.10 Indian Chemistry (IKS)

Philosophy of atom by Acharya Kanad: Acharya Kanad, an ancient Indian philosopher (founder of the Vaisheshika school), proposed that all matter is composed of indivisible particles called "**Paramanu**"

(atom), which combine to form larger particles — an early conceptual precursor to the modern atomic theory.

Unit 2.0 – Water

| 2.1 Introduction and Hardness of Water

Sources of water: Surface water (rivers, lakes) and groundwater (wells, bore-wells).

Hardness of water: The property of water due to which it does not form lather/foam readily with soap, caused by dissolved salts of calcium and magnesium.

- **Temporary hardness:** Caused by bicarbonates of Ca and Mg [$\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$]; removed by simple boiling.
- **Permanent hardness:** Caused by chlorides and sulphates of Ca and Mg (CaCl_2 , MgSO_4); not removed by boiling, requires chemical treatment.

| 2.2 Degree of Hardness and EDTA Method

Degree of hardness is expressed in terms of equivalent CaCO_3 (mg/L or ppm), since CaCO_3 has a convenient molecular weight (100).

EDTA Method: A complexometric titration method used to quantitatively determine the total hardness of water. EDTA (Ethylene Diamine Tetra Acetic acid) forms a stable complex with Ca^{2+} and Mg^{2+} ions in presence of Eriochrome Black-T indicator (wine red to blue colour change at the end point).

| 2.3 Municipal Water Supply and Softening Techniques

Treatment of water (Municipal supply): Involves screening, sedimentation, coagulation, filtration, and disinfection (chlorination) before supply.

Water Softening Techniques:

- **Soda-Lime Process:** Hardness-causing salts are converted to insoluble precipitates by adding calculated amounts of soda (Na_2CO_3) and lime [$\text{Ca}(\text{OH})_2$], which are then removed by sedimentation/filtration.
- **Zeolite (Permutit) Method:** Hard water is passed through a bed of sodium zeolite (Na_2Ze); $\text{Ca}^{2+}/\text{Mg}^{2+}$ ions are exchanged for Na^+ ions, softening the water. The exhausted zeolite is regenerated using NaCl brine solution.
- **Ion Exchange Method:** Hard water is passed through cation-exchange resin (removes $\text{Ca}^{2+}/\text{Mg}^{2+}$, releases H^+) followed by an anion-exchange resin (removes anions, releases OH^-), producing demineralized water.

| 2.4 Water Quality Index

BOD (Biological Oxygen Demand): The amount of dissolved oxygen required by microorganisms to biologically decompose organic matter present in water over a 5-day period at 20°C — indicates the degree of organic pollution.

COD (Chemical Oxygen Demand): The amount of oxygen required to chemically oxidize both biodegradable and non-biodegradable organic matter in water using a strong oxidizing agent (e.g., $\text{K}_2\text{Cr}_2\text{O}_7$).

Difference: COD value is always higher than BOD since it includes non-biodegradable matter too; COD test is faster (hours) compared to BOD (5 days).

Determination of Dissolved Oxygen: Done by the Winkler's (iodometric) method, where dissolved oxygen oxidizes $\text{Mn}(\text{OH})_2$ to a higher oxidation state, eventually liberating iodine which is titrated against sodium thiosulphate using starch indicator.

| 2.5 Indian Standard Specification of Drinking Water

As per BIS (IS 10500), drinking water should be colourless, odourless, free from harmful microorganisms, with pH between 6.5–8.5, total dissolved solids within prescribed limits, and hardness, fluoride, and heavy metal content within safe limits.

Unit 3.0 — Engineering Materials

3.1 Natural Occurrence of Metals

Minerals: Naturally occurring inorganic substances with a definite chemical composition.

Ores: Minerals from which metals can be extracted economically.

3.2 Metallurgy

Metallurgy: The science and technology of extracting metals from their ores and refining them for use.

- **Gangue:** Unwanted earthy/rocky impurities mixed with the ore.
- **Flux:** A substance added to combine with gangue to form a fusible slag.
- **Slag:** The fusible product formed by the reaction of flux and gangue, removed during smelting.

Steps in metallurgy: (1) Crushing and grinding of ore, (2) Concentration (removal of gangue), (3) Extraction (reduction to free metal — calcination/roasting + smelting/reduction), (4) Refining/purification (electrolytic refining, etc.).

3.3 Ancient Indian Metallurgy (IKS)

Ancient India had advanced metallurgical knowledge, exemplified by the **Iron Pillar of Delhi** (rust-resistant due to a protective phosphate layer) and the high-purity **Wootz steel** production, exported across the ancient world for sword-making.

3.4 Extraction of Aluminium, Iron and Copper

Aluminium: Extracted from **Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$)** ore by Bayer's process (purification) followed by **Hall-Hérault process** (electrolytic reduction of molten alumina dissolved in cryolite).

Iron: Extracted from **Haematite (Fe_2O_3)** ore in a **Blast Furnace** where coke reduces the ore to molten iron; limestone acts as flux to remove silica gangue as slag.

Copper: Extracted from **Copper pyrites (CuFeS_2)** by roasting, followed by smelting in a reverberatory furnace and Bessemerisation to obtain blister copper, finally purified by electrolytic refining.

3.5 Alloys

Alloy: A homogeneous mixture of two or more metals (or metal and non-metal), prepared to enhance properties like strength, hardness, corrosion resistance.

Purpose of alloying: To improve mechanical strength, hardness, corrosion resistance, and to lower melting point as required.

- **Ferrous alloys** (Iron-based): Example — Stainless Steel ($\text{Fe} + \text{Cr} + \text{Ni}$) — corrosion resistant, used in cutlery and surgical instruments.
- **Non-ferrous alloys:** Example — Brass ($\text{Cu} + \text{Zn}$) — used in utensils, fittings; Bronze ($\text{Cu} + \text{Sn}$) — used in statues, bearings.

3.6 Polymers

Polymer: A large molecule (macromolecule) formed by repeated linking of small repeating units called **monomers**.

- **Homopolymer:** Made from a single type of monomer. Example: Polythene (from ethylene).
- **Copolymer:** Made from two or more different monomers. Example: Nylon 66.
- **Natural polymers:** Occur in nature, e.g., cellulose, natural rubber, proteins.
- **Synthetic polymers:** Man-made, e.g., PVC, Nylon, Bakelite.

Addition polymerization: Monomers having double/triple bonds add together repeatedly without losing any atoms. Example: Polythene from ethylene.

Condensation polymerization: Monomers combine with the elimination of small molecules like water. Example: Nylon 66 from hexamethylenediamine and adipic acid.

Thermoplastic: Softens on heating and hardens on cooling; can be remoulded (e.g., Polythene, PVC).

Thermosetting plastic: Becomes permanently hard on heating (due to cross-linking) and cannot be remoulded (e.g., Bakelite).

3.7 Monomers and Applications

Polymer	Monomer	Application
Polythene	Ethylene	Bags, bottles, packaging
PVC	Vinyl chloride	Pipes, cables, flooring
Orlon	Acrylonitrile	Fabrics, sweaters
Terylene	Ethylene glycol + Terephthalic acid	Fabrics, fibres
Nylon 66	Hexamethylenediamine + Adipic acid	Ropes, fabrics, gears
Nylon 6	Caprolactam	Fibres, bristles
Bakelite	Phenol + Formaldehyde	Electrical switches, handles

3.8 Natural Rubber and Vulcanization

Natural Rubber: Obtained from latex of the rubber tree (*Hevea brasiliensis*); chemically a polymer of isoprene; soft, sticky, and loses elasticity at extreme temperatures in its raw form.

Vulcanization: The process of heating raw rubber with sulphur to form cross-links between rubber chains, improving elasticity, strength, and resistance to heat/abrasion.

Advantages of vulcanized rubber: Increased tensile strength, elasticity, durability, and resistance to temperature changes — used in tyres, conveyor belts, footwear.

Unit 4.0 — Solid State

4.1-4.2 Crystalline and Amorphous Solids

Crystalline solid: Has a definite, ordered, repeating three-dimensional arrangement of constituent particles; has a sharp melting point and is anisotropic (properties vary with direction). Example: NaCl, diamond.

Amorphous solid: Has no regular/ordered arrangement of particles; melts over a range of temperature and is isotropic. Example: Glass, rubber, plastics.

Classification of crystalline solids (based on binding forces):

Type	Constituent particles	Bonding	Example
Molecular	Molecules	Van der Waals/H-bond	Ice, Dry ice
Ionic	Ions	Ionic/electrostatic	NaCl
Metallic	Metal atoms/ions	Metallic bond	Cu, Fe
Covalent (Network)	Atoms	Covalent bond	Diamond, SiO ₂

4.3 Crystal Lattice and Unit Cells

Crystal lattice: A regular three-dimensional arrangement of points representing the positions of constituent particles in a crystal.

Unit cell: The smallest repeating unit of a crystal lattice that, when repeated in all directions, generates the entire crystal structure.

Types: - **Primitive (Simple) unit cell:** Particles only at the corners. - **BCC (Body-Centred Cubic):** Particles at corners plus one at the centre of the cell. - **FCC (Face-Centred Cubic):** Particles at corners plus one at the centre of each face.

4.4 Imperfections in Solids

Point defects are irregularities in the arrangement of constituent particles around a point/atom in a crystal:

- **Stoichiometric defects:** Do not disturb the overall stoichiometry of the crystal — e.g., Vacancy defect (missing particle), Interstitial defect (extra particle in interstitial space), Frenkel defect, Schottky defect.
- **Impurity defects:** Caused by the presence of foreign atoms in the lattice.
- **Non-stoichiometric defects:** Alter the ratio of constituents — Metal excess defect and Metal deficiency defect.

4.5 Electrical Properties — Band Theory

Band theory explains electrical conduction based on overlapping atomic orbitals forming energy bands: **valence band** (filled with electrons) and **conduction band** (empty/partially filled), separated by an **energy gap**.

- **Conductors (Metals):** Valence and conduction bands overlap — electrons move freely; low resistance.
- **Insulators:** Large energy gap between bands — electrons cannot jump to the conduction band.
- **Semiconductors:** Small energy gap — a few electrons can jump to the conduction band on slight heating/doping, giving intermediate conductivity (e.g., Silicon, Germanium).

4.6 Magnetic Properties

Property	Behaviour	Example
Ferromagnetism	Strongly attracted to magnetic field; retains magnetism	Fe, Co, Ni
Paramagnetism	Weakly attracted; no permanent magnetism	O ₂ , Cu ²⁺ salts
Diamagnetism	Weakly repelled by magnetic field	NaCl, H ₂ O
Anti-ferromagnetism	Magnetic moments cancel each other, net moment zero	MnO
Ferrimagnetism	Unequal magnetic moments align oppositely, net moment non-zero	Fe ₃ O ₄ (magnetite)

4.7 X-ray Diffraction — Bragg's Law

Bragg's Law relates the wavelength of X-rays, the spacing between crystal planes, and the diffraction angle, used to determine crystal structure: $n\lambda = 2d \sin\theta$ where n = order of diffraction, λ = wavelength of X-ray, d = interplanar spacing, θ = angle of incidence.

4.8 Melting Point Determination — Kjeldahl Method

The **Kjeldahl method** determines the melting point of a crystalline solid by filling a fine capillary tube with the powdered sample, attaching it to a thermometer, immersing both in a heating bath (e.g., Kjeldahl flask with liquid paraffin/sulphuric acid), and slowly heating while observing the temperature at which the solid just melts.

Unit 5.0 – Electrochemistry

5.1 Electrolytic and Metallic Conduction

Electrolyte: A substance that dissociates into ions in solution/molten state and conducts electricity (e.g., NaCl solution).

Non-electrolyte: A substance that does not dissociate into ions and does not conduct electricity (e.g., sugar solution).

Metallic conduction: Conduction of electricity by free electrons in metals; no chemical change occurs; decreases with rise in temperature.

Electrolytic conduction: Conduction by movement of ions in solution; accompanied by chemical change at electrodes; increases with rise in temperature.

Factors affecting electrolytic conductance: Nature of electrolyte, concentration, temperature, degree of dissociation, size/charge of ions.

5.2 Molar Conductivity and Kohlrausch's Law

Molar conductivity (Λ_m): Conductivity of a solution containing one mole of electrolyte, per unit volume.

Equivalent conductivity: Conductivity due to one gram-equivalent of electrolyte.

Variation of Molar conductivity: Molar conductivity increases with dilution due to increased degree of dissociation (for weak electrolytes) and reduced ionic interactions (for strong electrolytes).

Kohlrausch's Law: At infinite dilution, the molar conductivity of an electrolyte is the sum of the individual contributions of its constituent ions: $\Lambda^\circ_m = \lambda^\circ_{\text{cation}} + \lambda^\circ_{\text{anion}}$

5.3 Faraday's Laws of Electrolysis

First Law: The mass of substance deposited/liberated at an electrode is directly proportional to the quantity of electricity (charge) passed: $m = Z \cdot I \cdot t$ (Z = electrochemical equivalent, I = current, t = time)

Second Law: When the same quantity of electricity is passed through different electrolytes, the masses of substances deposited are proportional to their chemical equivalent weights.

Example: To find mass deposited: $m = (E \times I \times t) / 96500$, where E = equivalent weight, I = current in amperes, t = time in seconds.

5.4 Galvanic Cell and EMF

Galvanic (Voltaic) Cell: A device that converts chemical energy into electrical energy through spontaneous redox reactions, consisting of two electrodes (anode and cathode) dipped in electrolyte solutions, connected externally.

Electrode potential: The potential difference developed between an electrode and its solution.

Standard Hydrogen Electrode (SHE): A reference electrode with potential arbitrarily fixed at 0.00 V, used to measure standard electrode potentials of other electrodes.

EMF (Electromotive Force): The potential difference between the two electrodes of a cell when no current flows: $EMF = E(\text{cathode}) - E(\text{anode})$

Electrochemical series: Arrangement of elements in order of increasing/decreasing standard reduction potential, used to predict feasibility of redox reactions and relative reactivity of metals.

Nernst Equation: Relates electrode potential to ion concentration and temperature: $E = E^\circ - (RT/nF) \ln Q$
Used to calculate EMF at non-standard conditions of concentration, temperature, and pressure.

| 5.5 Batteries

Primary Cells (non-rechargeable): Example — **Dry cell** (Zn container as anode, carbon rod as cathode, $\text{NH}_4\text{Cl}/\text{MnO}_2$ paste as electrolyte) — used in torches, remotes.

Secondary Cells (rechargeable): Example — **Lead storage battery** (Pb anode, PbO_2 cathode, dilute H_2SO_4 electrolyte) — used in automobiles; can be recharged by reversing current flow.

Fuel cells: Convert chemical energy of fuels (e.g., H_2 and O_2) directly into electrical energy continuously, with water as the only by-product — used in spacecraft, eco-friendly vehicles.

| 5.6 Corrosion

Corrosion: The gradual destruction/deterioration of metals due to chemical or electrochemical reaction with their environment (oxygen, moisture, acids).

- **Dry corrosion:** Occurs through direct chemical reaction of metal with dry gases (O_2 , SO_2) at the surface, without involvement of moisture, forming an oxide layer.
- **Wet corrosion:** Occurs in the presence of moisture/electrolyte through an electrochemical mechanism (formation of anodic and cathodic areas), e.g., rusting of iron.

Prevention of corrosion: Painting/coating, galvanizing (zinc coating), electroplating, use of corrosion-resistant alloys, cathodic protection (sacrificial anode), application of inhibitors.